



DIGITAL TECHNOLOGIES

ZUKUNFT. DIGITAL. NACHHALTIG.

WiSe 2026

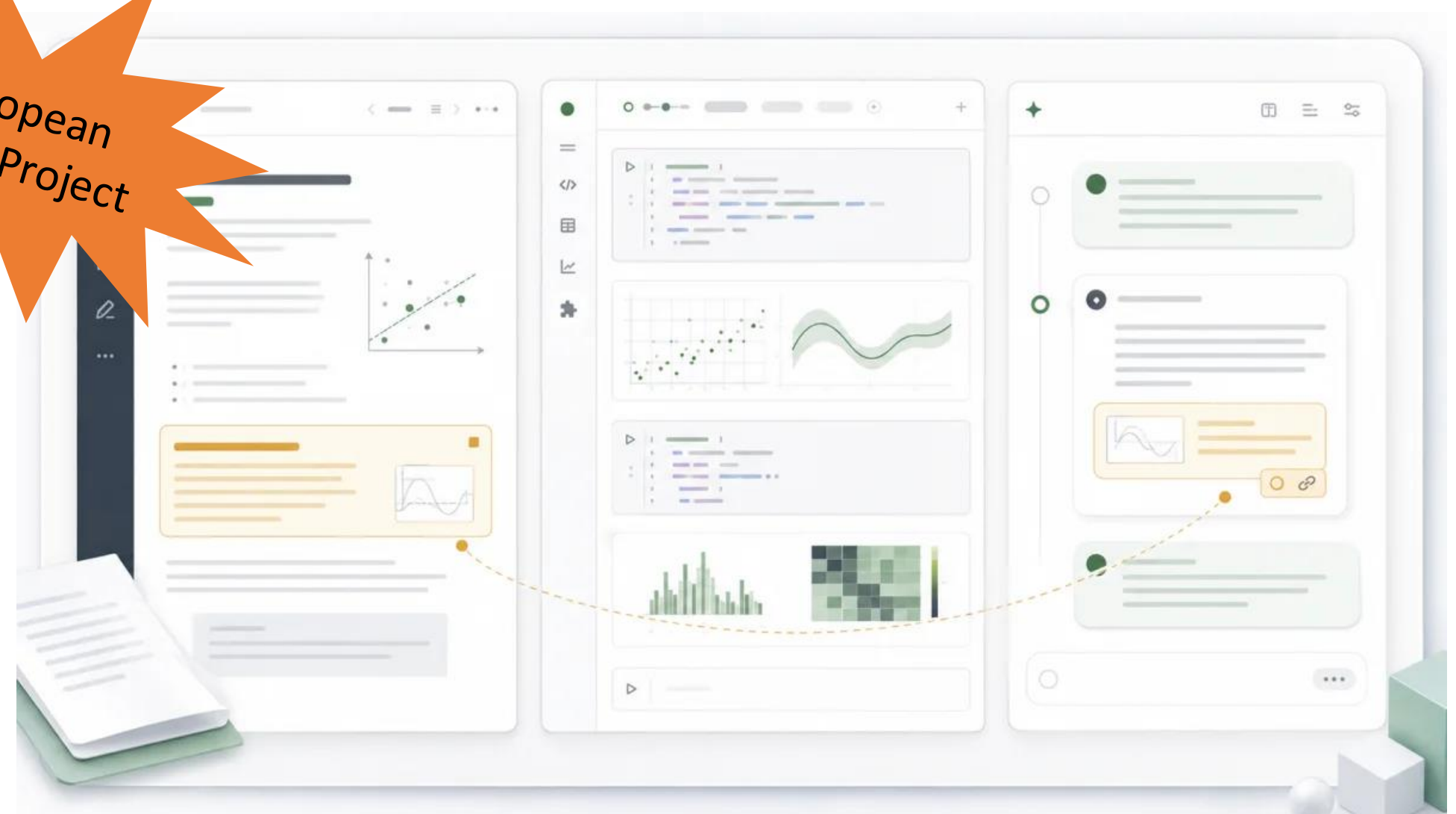
LECTRA

LLM-Enhanced Course Teaching and Retrieval Assistant

Machine Learning and COgnitive SoftwaRE (CORE)
Research Group
AI Team Projects (ATP)

Ein gemeinsames Studienprogramm der

European
Team Project



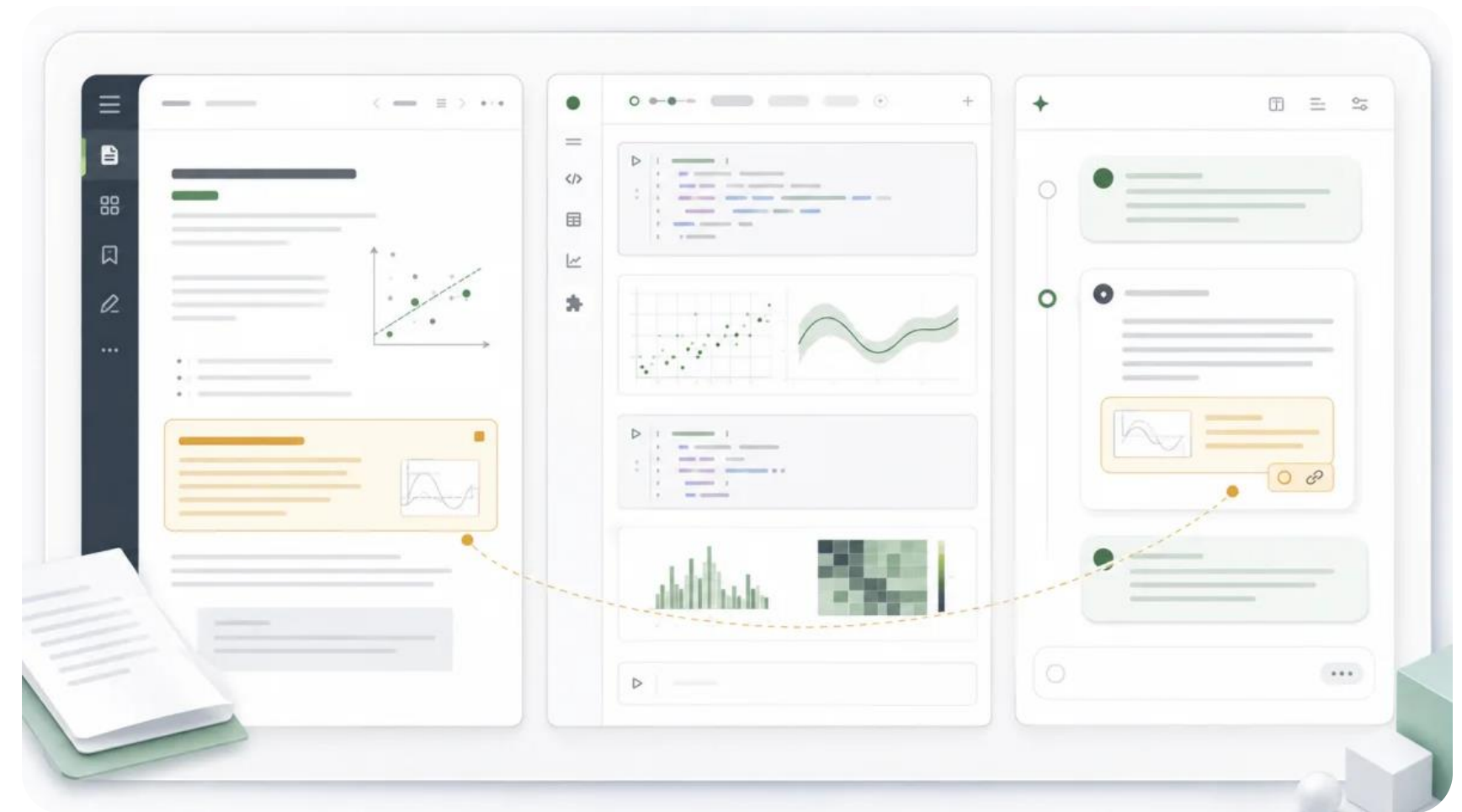
THE IDEA

Course material and an assistant in one workspace

LECTRA keeps

- lecture slides,
- PDFs,
- and rendered Jupyter notebooks

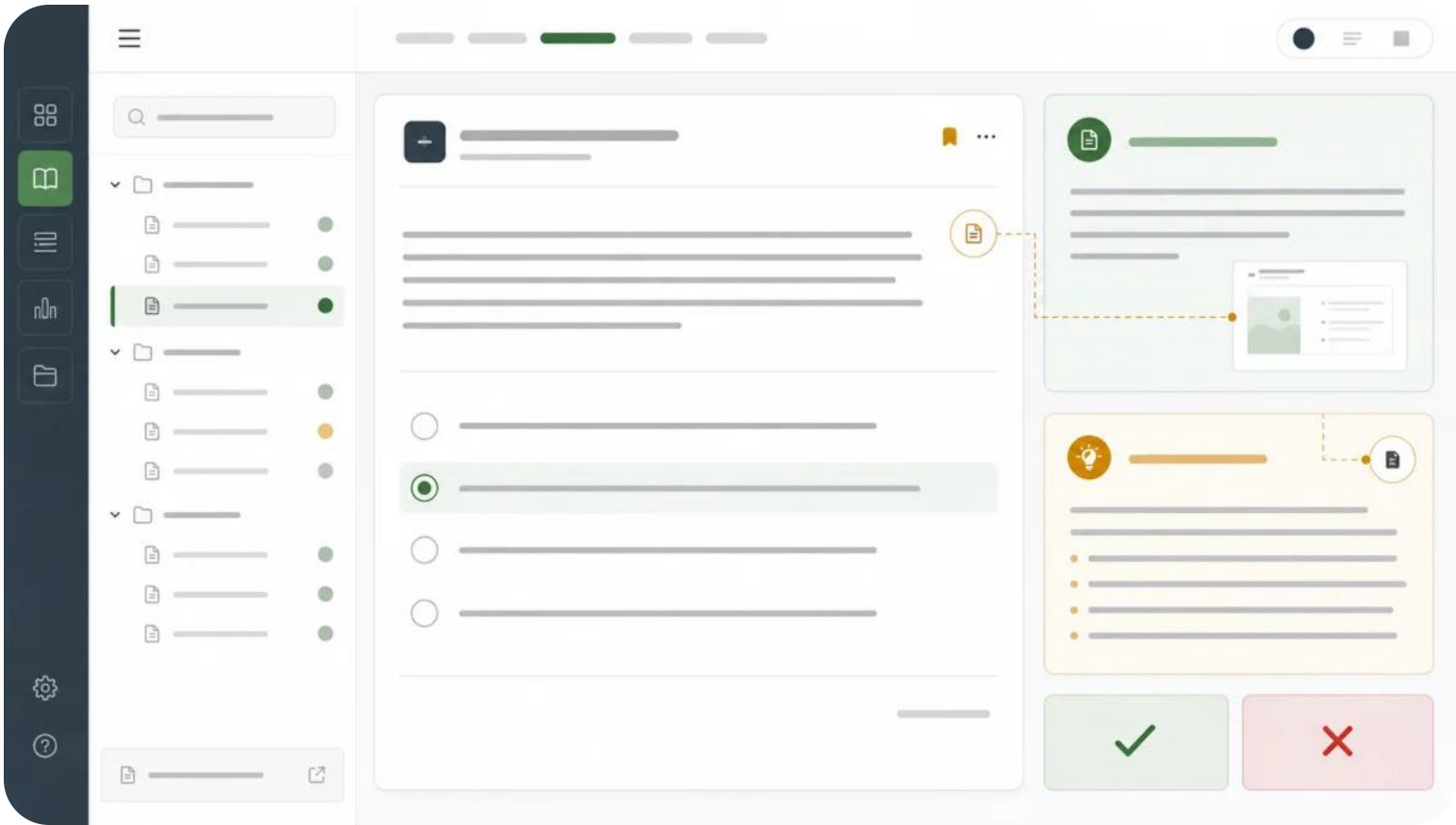
visible while students work with a context-aware LLM assistant to learn and deepen materials.



HOW IT COULD LOOK

A demonstrator session

Open	Lecture material and a related notebook side by side.
Ask	A question grounded in the selected slide, page, section, or code cell.
Explore	Source references, simpler explanations, examples, and notebook code.
Practise	An exam-style question, targeted feedback, and an adaptive follow-up recommendation.



ARCHITECTURAL OVERVIEW

What you are going to implement within the team



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Student

- Selects material
- Asks questions
- Receives explanations and recommendations



Student Learning State

- What the student knows
- Interactions, performance, gaps
- Personalized recommendations

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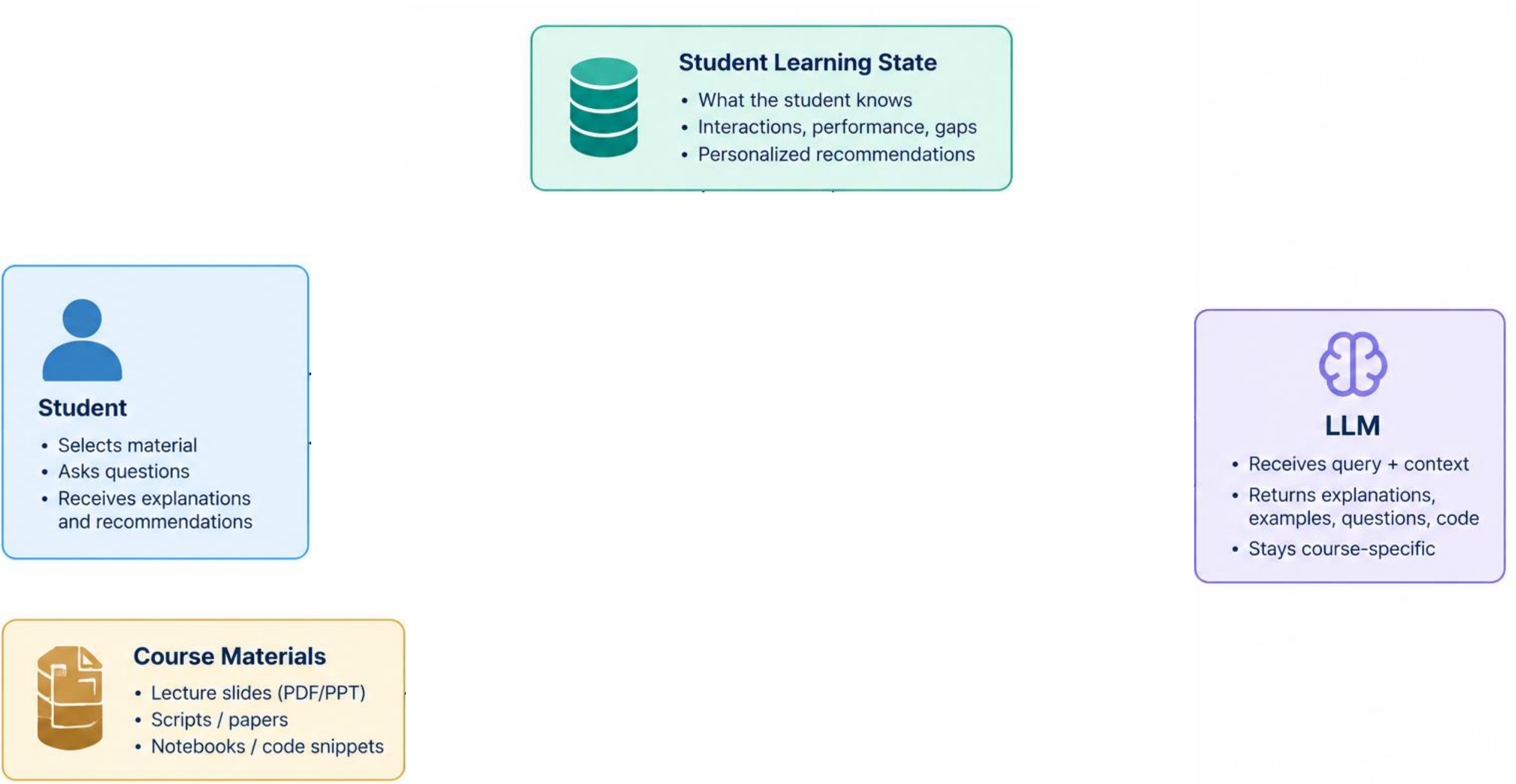


LLM

- Receives query + context
- Returns explanations, examples, questions, code
- Stays course-specific

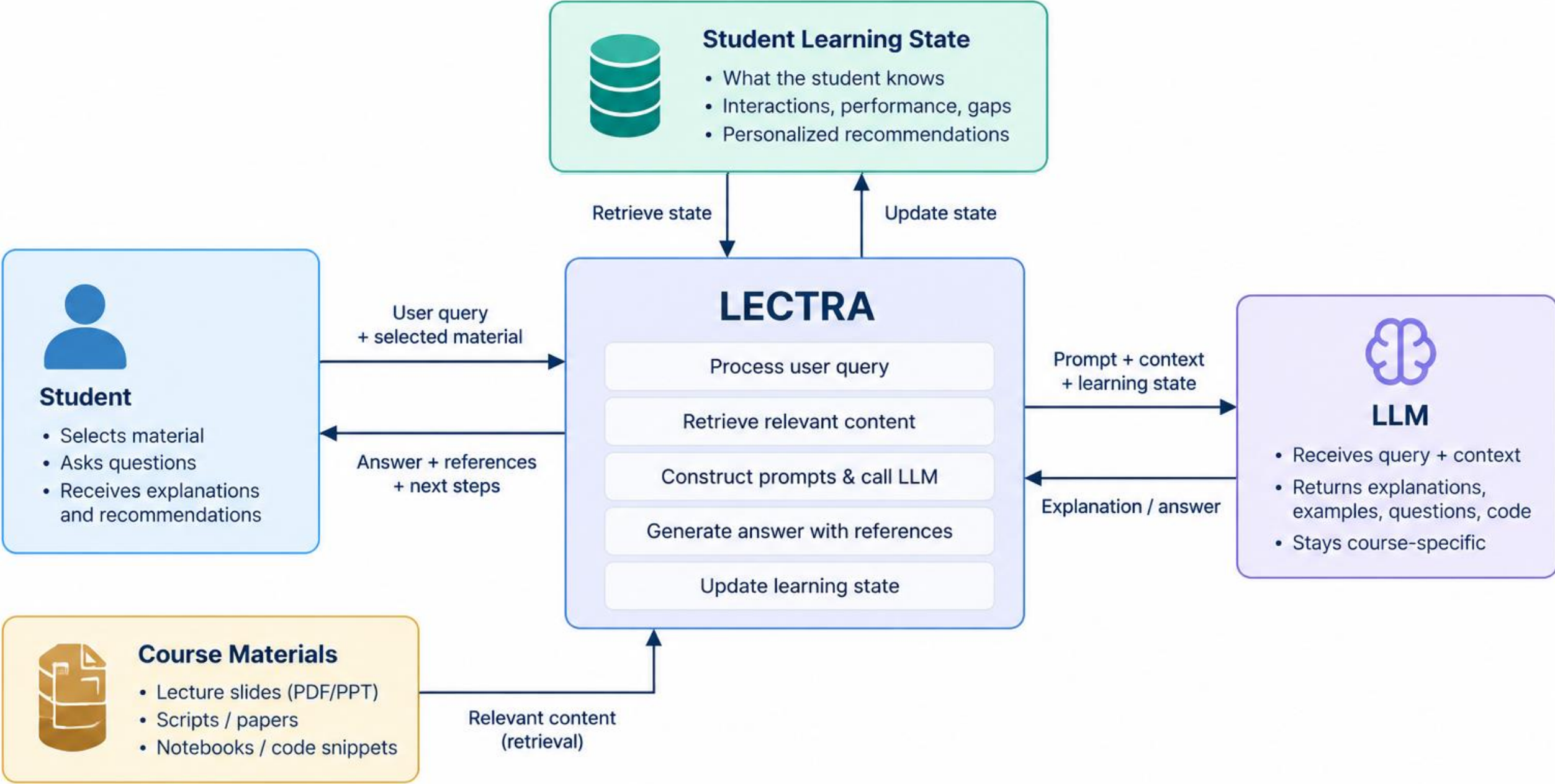
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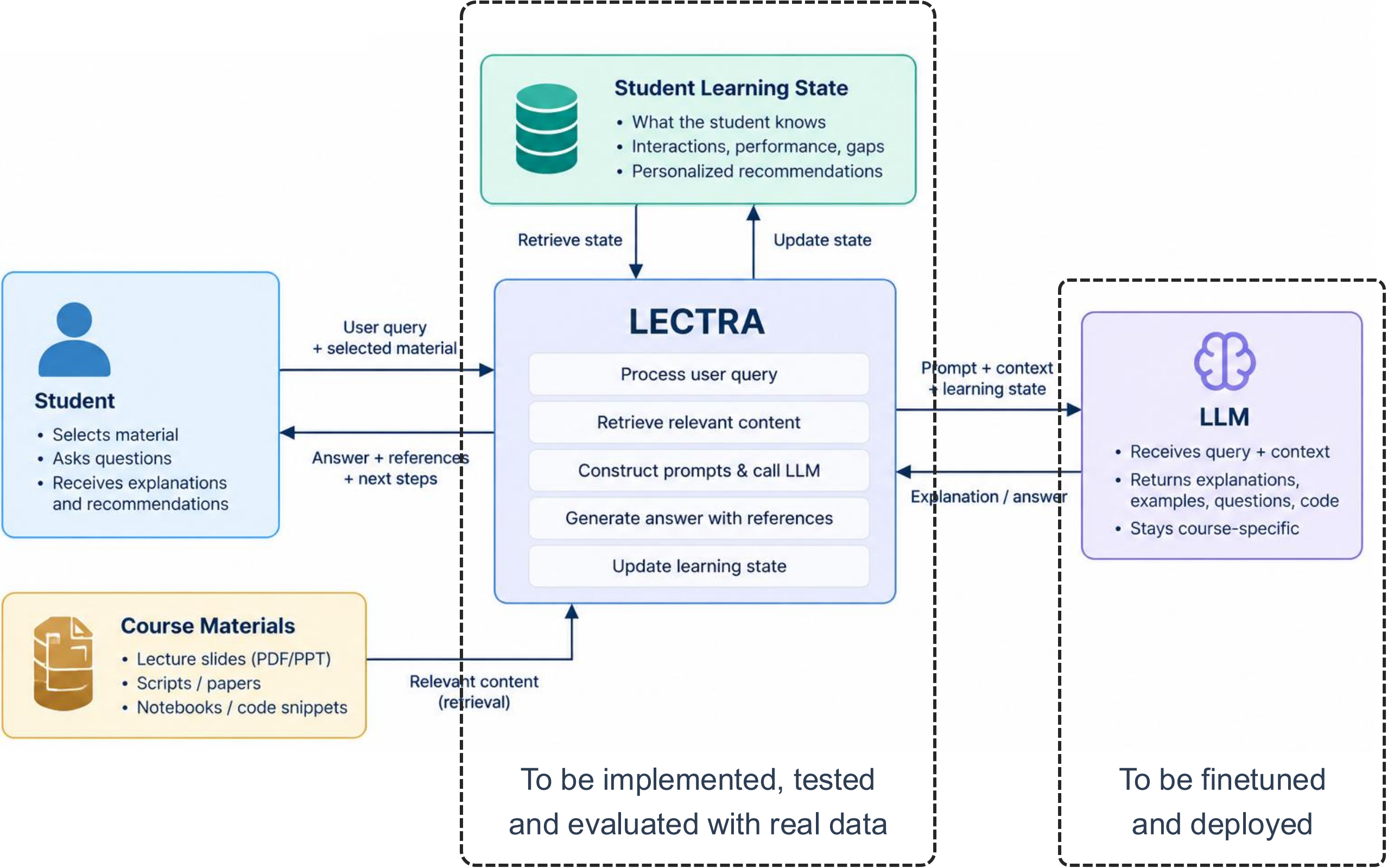
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Thinking about joining?

WHICH COMPETENCES ARE NEEDED?

- You are comfortable writing Python
- You are comfortable with git and working in a shared repository
- You enjoy working in a mixed international team across two universities
- Interest in machine learning helps – especially personalization and learning from user feedback
- Curiosity about how people learn, become uncertain, and benefit from feedback matters as much as model experience

WHAT DO YOU LEARN IN THIS PROJECT?

- Getting to know how LLMs work
- Building a platform that combines a PDF reader, with Jupyter notebook and an agent interface
- Fine-tuning a base model on one individual user's data
- Designing the architecture how LLMs work with PDF, code and user histories

Be part of an international team!

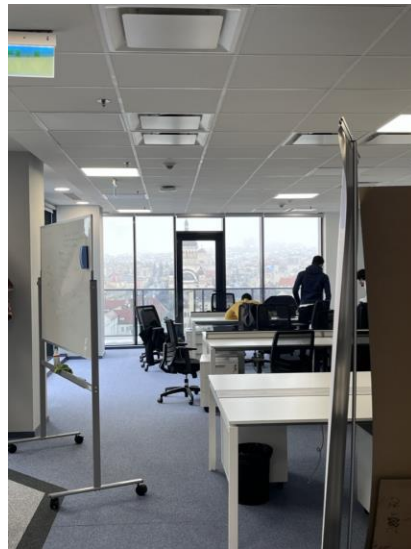
together with students from Babeş-Bolyai University, Cluj, Romania

For more info
and videos
visit:



<https://www.core-network.ai/ai-team-projects>

Intensive Week in Cluj



Social Events



SAVE THE DATE

Intensive Week Cluj:

~ 11.11. – 21.11.2026

Costs for flight and accommodation will be covered

Intensive Week Goslar:

~ 01.02. – 07.02.2027

CORE Projects in WiSe26

LECTRA

FastMile

Anbietende



UNIVERSITATEA
BABEȘ-BOLYAI

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